Exp - 8

22/09/2026

NMAP to discover live hosts using NMAP scans on TryHackMe Platform :-

Aim: To discover live hosts using NMAP scan (ARP, ICMP, TCP/UDP) on TryHackMe platform.

Procedure:-

Task-1 : Introduction :-

- ARP Scan : Uses ARP requests to discover live hosts.
- ICMP Scan : Send ping requests to identify live hosts.
- TCP/UDP ping scan : Sends packet to TCP/UDP ports to check host state.

Task-2 : Subnetworks :-

- Network Segment : Computer linked through common medium
- Subnetwork : Network segments linked together using same router (e.g., /16 or /24)

Task-3 : Target Specification :-

(i) List : nmap -sL <IP> scanme.nmap.org

(ii) Range : 10.11.12.15-20

(iii) Subnet : nmap -sL <machine-IP>/30

Task-4 : Discovering Live Hosts :-

- ARP used in local networks, ICMP in network layer, TCP/UDP in transport layer.

Task 5: NMAP host discovery using ARP:-

i) nmap -PR -sn <targets>

ii) nmap -PR -sn <machine-IP>/24

iii) nmap -PR -sn <attackbox>/24

Task 6: NMAP host discovery using ICMP:-

- nmap -PE -sn <target> (Echo)
- nmap -PP -sn <target> (timestamp)

Task 7: NMap host discovery using TCP/UDP:-

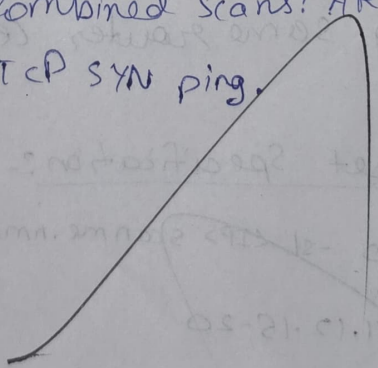
- nmap -PS -sn <target> (TCP SYN)
- nmap -PS21-25 -sn <target> (Port range)
- nmap -PA -sn <target> (TCP ACK)
- nmap -PU -sn <target> (UDP ping)

Task 8: Using Reverse-DNS lookup:-

- nmap -R <target> queries DNS for all hosts.

Task 9: Final Task:-

- combined scans: ARP, ICMP echo/timestamp, and TCP SYN ping.



Result:-

The experiment to discover live hosts using Nmap was carried out successfully.

22/1/24